

Article

Standing Strong: Simulation Training and the Emotional Resilience of Healthcare Providers During COVID-19

Alice Yip ^{1,*}, Jeff Yip ² , Zoe Tsui ¹ and Jacky Chun-Kit Chan ³ ¹ S.K. Yee School of Health Sciences, Saint Francis University, 2 Chui Ling Lane, Tseung Kwan O, New Territories, Hong Kong, China; ztsui@sfu.edu.hk² Hong Kong Institute of Paramedicine, Hong Kong, China; jeffreyyip@gmail.com³ School of Nursing and Health Sciences, Hong Kong Metropolitan University, Hong Kong, China; jckchan@hkmu.edu.hk

* Correspondence: aliceyip138@gmail.com

Abstract: This qualitative study explored the development of an innovative simulation training program designed to bolster the preparedness and resilience of healthcare teams during the unprecedented challenges of the COVID-19 pandemic. Focus groups with interprofessional clinicians illuminated key educational priorities, revealing a need for enhanced safety protocols, clear in communication, and targeted training to address knowledge gaps specific to the novel coronavirus. Recognizing the profound emotional toll of the pandemic, the program also emphasized cultivating compassion and fostering emotional resilience alongside the essential clinical skills. By immersing participants in realistic, evolving scenarios that were reflective of the dynamic COVID-19 landscape, the simulations offered a safe space to rehearse critical skills, practice crisis resource management, and build confidence in navigating the complexities of pandemic care. This tailored approach aimed to empower healthcare teams not only with enhanced knowledge and expertise but also with the emotional fortitude and resilience necessary to provide optimal patient care while safeguarding their own well-being throughout the ongoing COVID-19 pandemic. The findings highlight the profound potential of simulation-based training to strengthen both individual and systemic resilience within healthcare systems facing the enduring strain of this global health crisis.



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1. Introduction

As the COVID-19 pandemic emerged, our limited understanding of how the virus spread placed healthcare professionals at a significant risk of nosocomial infection [1,2]. During the early stages of the pandemic, global infection rates among healthcare providers were alarmingly high [3,4]. The resulting psychological fatigue and trauma-related stress impacted both their emotional well-being and cognitive functions, potentially affecting clinical judgement and increasing the risks exposed to patients [5,6]. The pandemic exposed critical limitations regarding healthcare professionals' preparedness for novel situations, particularly those involving critically ill patients. This emphasized the necessity of training programs that address a wider range of competencies beyond clinical expertise. Some research in the literature emphasizes the importance of non-technical skills, such as effective communication, in delivering holistic patient care and enhancing our response

to pandemics [7–9]. By cultivating these skills, healthcare providers could improve their capacity to navigate uncharted territory.

The strategies employed by healthcare providers in response to direct threats to their personal safety during care delivery are subject to various cognitive and psychological determinants. The profound impact of widespread health crises, such as the recent pandemic, has illuminated the complex array of challenges that healthcare professionals confront. These challenges are not unchanging; they span technical, cognitive, and psychological dimensions, each demanding specific attention and preparedness [10–15]. To effectively equip healthcare providers for such demanding realities, educational strategies must evolve beyond traditional methods. Clinical simulation has emerged as a particularly potent tool in this context, offering a dynamic and safe environment in which to explore these multifaceted demands. Through carefully designed scenarios and structured debriefing, simulation enables healthcare providers to engage more thoroughly with cognitive processes, such as decision-making under pressure and situational awareness, and enhances the psychological resilience required in high-pressure scenarios. While the acquisition of technical skills for specific tasks and operations remains essential, it is this holistic approach, addressing the interwoven cognitive and psychological aspects of safety, that is important. Understanding and cultivating these dimensions through practical and introspective methods such as clinical simulation is therefore crucial [16,17]; this cannot only hone procedural competence but, more critically, improve the well-being of providers and safeguard optimal patient care when facing the inevitable stressors of challenging healthcare situations. In contrast to technical skills, non-technical skills encompass the behavioral adaptations needed for diverse clinical situations. These skills, including interpersonal relationships and cognitive functions, enhance the effectiveness of practical and clinical competencies. Non-technical training can encompass areas such as communication, teamwork, emotional regulation, leadership, situational awareness, and decision-making. The power of simulation-based learning to provide both experience and opportunities for reflection has made it a vital and growing aspect of healthcare education [18,19]. However, considerable attention should be devoted to its efficacy in ensuring the safety of healthcare providers and, most significantly, patients [20–23].

Healthcare teams demonstrated remarkable resilience throughout the COVID-19 pandemic [24]. Faced with unprecedented challenges, they adapted to novel learning platforms and techniques in order to acquire critical skills, such as the safe and effective use of personal protective equipment (PPE) [25,26]. This knowledge provided greater comfort and reduced stress when caring for patients, particularly when managing airways, which is a high-risk procedure. Their adaptability and commitment to ongoing learning were essential to ensuring the safety of both patients and healthcare providers during this extraordinary time [27–31].

Theoretical Framework

The COVID-19 pandemic placed unprecedented stress on healthcare providers, impacting their well-being and potentially compromising patient care. Stress Inoculation Training (SIT), a cognitive-behavioral technique, offers a structured approach to building resilience in such challenging circumstances [32]. When integrated with simulation-based learning, SIT becomes even more effective. The training involves three key phases: understanding the nature of stress; acquiring coping skills, such as cognitive restructuring and relaxation techniques; and applying these skills in realistic simulated pandemic scenarios. Simulation provides a safe space in which to practice communication strategies, manage critical situations, and build confidence. This combined approach aims to reduce stress

and anxiety among healthcare providers, enhance job satisfaction, prevent burnout, and ultimately improve the quality and safety of patient care.

Furthermore, training programs can utilize simulated conditions that closely approximate actual work environments to engage healthcare providers from multiple disciplines. These sessions can focus on cases that are directly relevant to their specific areas of practice, as demonstrated in this research. The use of clinical simulation as an educational approach in training healthcare providers has a long track record of positive outcomes [19,33–36]. While the need for effective simulation training during pandemics is widely acknowledged, there remains an urgent need for further scholarly inquiry [37–44]. Many existing studies focus on quantitative analyses of course structure or program outcomes. However, there is a significant gap in qualitative research exploring the specific factors that contribute to successful simulation-based training, particularly in the context of a pandemic. This study addresses a gap in the literature by employing a qualitative methodology to delineate the critical factors strengthening the success of simulation-based training programs. These programs are tailored to multidisciplinary healthcare teams responsible for the care of patients with infectious diseases. The investigation specifically explores how effective training programs incorporate strategies that equip healthcare providers with the skills required to manage stress in unpredictable and demanding environments, such as those precipitated by pandemic crises such as the COVID-19 pandemic. Additionally, this research examines how these initiatives foster physical and psychological resilience among these professionals during such public health emergencies. The aim of this study is to develop a tailored, simulation-based training program informed by the lived experiences and insights of healthcare providers, ultimately enhancing our preparedness for further pandemic responses.

2. Methods

2.1. Study Design

This qualitative study used semi-structured focus groups to assess the preparedness and needs of healthcare professionals during the pandemic. Insights from three focus groups were analyzed to identify key skills and create relevant simulation scenarios. The descriptive phenomenological approach and flexible interviews provided a comprehensive understanding of experiences. Reporting adhered to Tong et al.'s (2007) criteria for qualitative research, ensuring rigor and validity [45].

2.2. Participants

Participants for the simulations were recruited at a local conference on COVID-19 management, engaging practicing physicians and nurses. Patient care assistants were primarily sourced through nursing staff social networks utilizing snowball sampling. Participants needed to demonstrate a willingness to engage in simulations without prior knowledge of the scenario and to have sustained at least one year of continuous clinical practice in an acute care setting.

2.3. Data Collection

Data were collected through three Zoom focus groups conducted in Hong Kong between June 2022 and September 2022, with data saturation achieved. The first group comprised five physicians and one associate professor actively involved in clinical practice and research. The second group consisted of six specialized nurses from diverse practice areas. The third group included six patient care assistants with extensive experience in acute care setting, primarily in general medicine and infectious disease wards. Two experienced facilitators (J.Y. and A.Y.) guided the discussions, which lasted approximately 103 min each;

a semi-structured guide (Supplementary Table S1) was used to elicit participant perceptions, reflections, self-assessed responses, care improvements, and needs. Field notes were taken throughout each focus-group session. Data saturation was achieved when further data collection yielded no new insights, themes, or concepts relevant to this study. This point was reached after the third group session, as the narratives of the final two participants introduced no novel information. With no participants withdrawing, data collection was subsequently concluded.

2.4. Analysis Strategy

This study employed content analysis, a powerful methodology used to extract unique insights and deepen our understanding of the phenomena observed, ultimately illuminating their practical applications [46,47]. After the data-collection phase, NVivo software (NVivo Version 12, QSR International, Burlington, MA, USA) was utilized for the timely analysis of interview transcripts. Interviews were transcribed verbatim to accurately capture participant dialogues. Each transcript was subjected to multiple close readings to ensure that they were thoroughly understood and then deconstructed into condensed meaning units. These units were systematically abstracted and coded. Similarities and differences in the codes facilitated their methodical classification into sub-categories and broad themes. The final phase of analysis involved a continuous process of comparing and reflecting on the identified sub-categories [46]. This iterative approach facilitated the revelation of implicit content within the data, and it finally led to the emergence of the study's primary theme. To ensure linguistic accuracy, bilingual experts fluent in English translated the final analysis. The coding process, led by the first author (A.Y.) as the primary coder, involved presenting preliminary findings at scheduled team meetings. These sessions facilitated feedback on coding and categorization, ultimately leading to a team consensus and the finalization of the codebook.

2.5. Rigor

To ensure the integrity of the study, a rigorous methodological framework, as outlined by Lincoln and Guba (1985) [48], was diligently followed. Trustworthiness was reinforced through a validation process comprising the post-interview checking of members, a data analysis audit, a reflexive discussion on the participants' psychological feelings by the research team, and the snowball sampling of healthcare providers for wide-ranging perspectives and experiences. In this process, the participants confirmed the accuracy of their responses, as detailed in (Supplementary Table S2). This approach enables us to guarantee that the findings genuinely reflect the experiences and perspectives of the participants, thus minimizing potential research bias.

2.6. Ethical Considerations

This study adhered to the ethical principles outlined in the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of the clinical simulation training center prior to the study's commencement (HRE220122). All participants received comprehensive information about the nature and objectives of the research and provided voluntary informed consent, documented through signed consent forms.

3. Results

The study involved 18 healthcare providers, whose characteristics are delineated in Table 1. An analysis of the data revealed three primary themes related to educational needs. These themes, which are elaborated upon in Table 2, were integrated into the training program's framework to enhance its pedagogical effectiveness:

Table 1. Participant characteristics.

	Frequency (n)	(%)
Occupation		
○ Physician	6	33.3
○ Non-physician (includes registered nurse and patient care assistant)	18	66.7
Highest education qualification		
○ ≥Master level	8	44.5
○ Bachelor level	4	22.2
○ Diploma level	6	33.3
Clinical experience in acute care hospital settings		
○ <10 years	10	55.6
○ ≥10 years	8	44.4

Table 2. Themes of the study.

Themes	Description	Sample Codes
Seeking clarity and safety through pandemic training	Focus groups stress clear pandemic training for improved patient care	As a frontline doctor, I'm overwhelmed by this alarming pandemic. Our routines are upended, and medical practices have rapidly adapted for confirmed cases. This constant change types of unsettling uncertainty, as protocols and clinical guidance seem lost before reaching use. Like many, I yearn for clarity and guidance in these challenging times. (Physician 03)
		Collecting patient samples is daunting and carriers' immense responsibility. Without proper, secure training, overlooking vital infection control steps is easy. Risking viral exposure a chilling prospect. Thus, reflective practice sessions are vital. They provide a structured approach to identify weaknesses, implement improvements, acknowledge our vulnerability, and motivate striving for enhanced safety measures. (Nurse 01)
		Simulation exercises are invaluable, but more time and video access would be enormously beneficial. The current 45 min feels rushed; an extra 15–30 min would allow deeper engagement, full video viewing, and thorough debriefing to solidify understanding. This realistic training builds confidence and safety exposes vulnerabilities. Unlike written test, simulations challenge critical thinking and emotional intelligence essential for quality care. Extended training creates more seasoned, capable, and compassionate nurses, empowering sound decisions in high-stress situations. This investment directly benefits patients and is worth the extra time. I hope this extension is granted. (Nurse 02)
Closing knowledge gaps and strengthening protocols through immersive simulation	Clear, aligned objectives and checklists vital for training	I discovered a distinct fondness for this evaluative scale (Likert-type scale). It's not just about its uncomplicated nature, it's about how it encapsulates our complex roles. It goes beyond clinical skills, reaching into our ability to connect and empathize with patients. This scale seems to capture the essence of our work. It beautifully reflects a blend of science and understanding in a simple measure. (Physician 04)
	Healthcare professionals stress the significance of custom training simulations	The evaluation scale is more than a tool; it's a metric of our clinical skills in managing infected patients, gauging our preparedness and proficiency. Its application should be straightforward, like a compass, not a puzzle. Knowing specialists developed it instills confidence, validating its credibility to objectively assess abilities and guide us to become more effective caregivers. (Physicians 06)
		The COVID-19 pandemic enhances the need for effective practical training in high-risk procedures like suctioning or tracheostomy care. Visualization is insufficient; mastering PPE (personal protective equipment) and infection control requires hands-on practice. Conceptual understanding must be paired with tangible experience and feedback to cultivate the skills and confidence needed for real-world critical tasks. (Nurse 04)

Table 2. *Cont.*

Themes	Description	Sample Codes
		I have come to appreciate the value of training tailored to the specific needs of our workplace. Effective training should extend beyond theoretical concepts and focus on addressing common challenges and procedures encountered in our daily routines. By prioritizing relevant, participatory training, we can ensure that acquired knowledge is readily retained and translated into instinctive action when faced with actual scenarios. (Patient care assistant 02) Participating in the recent infection simulation training was invaluable. Beyond new knowledge, it offered critical insights for improving my safe patient care practices. Reviewing my video performance with a checklist allowed thorough self-assessment, pinpointing potential infection spread risks, like equipment adjustments. Observing my performance revealed subtle errors I'd have otherwise missed. The training deconstructed complex workflows into manageable steps, significantly boosting my confidence. As a frontline worker, this renewed my assurance in protecting myself, colleagues, and patients. Experiential learning cultivated both competence and confidence through hands-on practice and reflective analysis. (Patient care assistant 04)
Bolstering knowledge and compassion in infection care	Infection education is essential for informed, safer healthcare practices	Managing patient care, I yearn to better understand infectious disease patients' holistic needs—physical, psychological, and emotional. When their condition deteriorates, offering comfort and reassurance is supreme, demanding concern and empathy beyond medical expertise. I found this sentiment particularly powerful, as it not only highlights the immense emotional labour involved but also emphasizes our articulated need for comprehensive training, to better equip us as pillars of strength and sources of hope for our most vulnerable patients during such unprecedented times. (Nurse 03) Though aware of typical infection symptoms like shortness of breath and persistent coughing, this knowledge alone is insufficient for COVID-19. This novel virus has unprecedentedly reshaped all aspects of healthcare. We grapple with understanding the virus and its extensive effects on daily practice and patient management. Beyond simple information, a clear framework is essential to cover patient care in this new infectious disease landscape, guiding us through uncharted territory. (Patient care assistant 05) Participating as an isolated standardize patient offered significant insights into patient and healthcare provider challenges. Experiencing simulated prolonged isolated fostered deep empathy for its physical and emotional duty. This highlighted previously unconsidered needs, prompting a redesign of my care approach with greater feeling and patient-centeredness. Stepping into both roles cultivated heightened sensitivity to their realities. This empathy profoundly impacted my perspective, reinforcing the importance of humane and compassionate care addressing holistic needs. It was an unforgettable reminder that compassionate understanding and human connection are as essential as medical expertise. (Nurse 06)

3.1. Theme 1: Seeking Clarity and Safety Through Pandemic Training

The participants in this study represented diverse professional roles and levels of educational attainment. Despite these differences, all groups emphasized the need for clear, up-to-date guidance and a strong theoretical foundation regarding our preparedness for pandemics. This included a thorough understanding of viral transmission mechanisms, as such knowledge can inform the clinical decision-making processes of healthcare professionals when caring for patients with infectious diseases. Notably, participants across all groups expressed that more time should be dedicated to debriefing sessions. This finding suggests that extended debriefing periods may foster a sense of psychological safety, regardless of individual competency levels. While Likert-scale evaluations may indicate clinical competence, the participants highlighted that feelings of insecurity can persist. Therefore, allocating ample time for debriefing may be crucial in promoting psychological safety and addressing individual concerns.

3.2. Theme 2: Closing Knowledge Gaps and Strengthening Protocols Through Immersive Simulation

Physicians and nurses emphasized the importance of establishing clear, specific, and measurable learning objectives for each simulation training scenario. These pedagogical goals must be effectively communicated to all participants prior to the simulation exercise.

The accompanying evaluation checklist should demonstrate internal consistency and direct alignment with the intended learning objectives. To ensure a comprehensive and accurate assessment, the checklist should specifically detail the expected actions of participants. Furthermore, incorporating items that allow for a nuanced evaluation using a Likert scale is highly recommended. This approach facilitates a more detailed and insightful analysis of participant performance, moving beyond simple binary assessments.

Beyond cultivating a wide range of technical skills, the participants consistently emphasized the importance of regular refreshers and updates regarding the proper utilization of personal protective equipment (PPE). They believe that incorporating customized, scenario-based simulations is the most effective method for achieving and maintaining proficiency in these critical safety practices.

Video-assisted debriefing represented an impactful tool, enabling participants to critically analyze their performance and pinpoint areas in which workflows could be optimized. The visual and self-reflective nature of video debriefing provided significantly greater in-sight than verbal discussions alone, allowing participants to readily identify and address performance gaps.

3.3. Theme 3: Bolstering Knowledge and Compassion in Infection Care

There is a strong consensus among physicians that a comprehensive understanding of the pandemic, including its transmission dynamics and clinical presentation, is crucial for all healthcare team members, including patient care assistants. This perspective is shared by patient care assistants themselves, who believe that possessing a strong knowledge base would positively influence their practices and highlight the importance of integrating stringent infection control measures into their daily routines. Furthermore, the collective experience of participants emphasized the need to re-evaluate and re-design care plans to incorporate more compassionate and holistic elements.

4. Discussion

This study aimed to develop a customized simulation-based training program designed to strengthen the preparedness of multidisciplinary healthcare teams for a pandemic. The findings highlight key opportunities to enhance readiness by addressing critical knowledge gaps, promoting adherence to essential agreements, and cultivating compassionate care competencies.

The clear need for understandable guidelines and a strong comprehension of transmission dynamics emphasized the importance of knowledge concerning the pandemic. This foundation is essential for developing and implementing evidence-based practices. A strong understanding empowers healthcare providers to confidently navigate the uncertainties of pandemic care, echoing findings that emphasize the value of conceptual clarity in such situations [5,39]. Targeted education can bridge knowledge gaps and provide a clear rationale for decision-making, ultimately leading to more effective responses.

The need for improved PPE techniques highlights the importance of hands-on training in ensuring proper compliance. As contexts rapidly evolve, ongoing simulations offer a safe environment for healthcare providers to refine their skills and maintain proficiency, even as guidelines are updated. Simulation-based education effectively bridges this gap by allowing healthcare professionals to translate updated rules into proficient practice [37,38]. This approach uniquely combines knowledge acquisition with experiential mastery, enabling repetitive, risk-free rehearsals.

Incorporating standardized patients into simulation training provided invaluable opportunities for healthcare providers to gain direct insight into the patient experience [49]. This experiential learning fosters a deeper understanding of patient needs, motivating

providers to collaborate and provide more holistic, patient-centered care. The training also cultivated a sense of psychological safety, a dimension that is difficult to quantify through traditional evaluation methods such as Likert scales [50]. By temporarily stepping into the shoes of isolated patients, healthcare providers developed heightened empathy and compassion, informing more humanistic care practices. This emphasizes the value of integrating both provider and standardized patient perspectives into future simulation curricula. Qualitative feedback suggests that this hands-on approach offers nuanced opportunities for improving care that quantitative outcome measures alone may not.

Cultivating compassion and humanistic care is equally crucial. Technical skills alone cannot foster genuine care; true caring stems from connecting with patients' deep-seated psychosocial and emotional needs. Education must move beyond rote skills and prioritize patient humanity, nurturing empathy in healthcare providers [4,6]. With compassion as its cornerstone, care transcends mere technical proficiency, becoming a holistic practice that embraces the patient's mind, body, and spirit.

These findings highlight the potential of simulation training to deepen conceptual understanding, enhance observation, and foster compassion. As healthcare providers navigate the ever-changing landscape of infectious diseases, empowering them through timely knowledge transfer, skills development, and compassion cultivation could strengthen both the care they provide and their own resilience [40,43]. These findings highlight the need for comprehensive training curricula that integrate knowledge, skills, and compassionate care. Simulations offer a safe space for healthcare professionals to hone these skills through hands-on practice, while aligned assessments validate knowledge acquisition. Together, these elements foster both the confidence and competence needed for effective clinical practice.

Limitations

As this is an initial qualitative study, the findings may not be generalizable beyond the specific participants and context. A more exhaustive exploration of alternative interpretations regarding the efficacy of simulation-based training and a deeper analysis of its inherent challenges in fostering resilience during the COVID-19 pandemic would enhance the comprehensiveness of this study. Furthermore, the findings are subject to constraints, including potential selection bias due to the reliance on voluntary participants; the context-dependency of qualitative data, which may influence transferability; and the inherent difficulties in objectively measuring complex psychological constructs such as empathy or perceived psychological safety within the simulation environment.

Further research should employ vigorous research designs to rigorously evaluate the proposed training program. For example, comparing virtual and in-person modalities may reveal unique advantages regarding the development of competence in each format. In addition, opportunities to expand the training curricula and assess their impact across diverse healthcare roles, teams, and organizational settings also exist.

5. Conclusions

This study demonstrates the value of customized simulation training in meeting the diverse needs of frontline teams during a pandemic. Simulation provides a unique platform for rehearsing new arrangements and care workflows, effectively bridging the gap between guidelines and proficient practice. Importantly, the training fosters psychological safety and surpasses typical skills assessments. Reviewing full video recordings offered deeper insights into performance gaps than standard debriefing conversations. Taking on the roles of both healthcare providers and standardized patients fostered empathy and motivation in the delivery of patient-centered care. As healthcare systems navigate the evolving landscape of the pandemic, adaptable simulation training strengthens expertise, reinforces rules, and

cultivates compassion in order to bolster the resilience of teams. The findings of this study are not only individual; they emphasize a critical need for the participants' working institutions, particularly at the management level, to cultivate a deeper understanding of their staff's situation and actively seek improvements. Building on this, future research should not only evaluate the impact of expanded simulation curricula on the quality of care, provider well-being, and system-wide responsiveness during both crisis response and recovery phases but also explore these institutions and commitments in order to optimize the benefits of such training initiatives.

Ultimately, these findings advocate for simulation as a powerful tool for integrating knowledge, skills, reflective practice, and humanistic values to cultivate more compassionate and collaborative healthcare.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/covid5060092/s1>, Table S1: Interview guide questions; Table S2: Technique to maximize rigor.

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Informed Consent Statement: Informed consent was taken from participants to participate in the study. In addition, all participants were assured that their shared experience and interview content would be reported in international journals anonymously.

Data Availability Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Conflicts of Interest: The authors declare no conflicts of interest.

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